

Azure Databricks ETL Project

E-commerce/Retail Data Platform — Setup & Roadmap Guide

1. Your Environment (Task 0 results)

Item	Value / Status
Azure subscription	Existing subscription (already have one)
Databricks workspace	Existing workspace, reused for this project
Pricing tier	Trial (Premium - 14-Days Free DBUs)
Region	Central India
Admin access	Yes — Admin Console is now part of Settings
Unity Catalog / catalogs / storage / compute	Still to confirm — see Section 2 note

Premium tier means Unity Catalog and serverless compute should both be available to you. We still need to confirm the remaining Task 0 checks (Unity Catalog enabled, which catalogs you can see, whether serverless shows up on the compute-creation screen, and any existing storage account in your own resource group) once you're ready to look — this guide will be updated as those come in.

2. Critical: How Your Trial Actually Bills

"Trial (Premium - 14-Days Free DBUs)" only waives the **Databricks Unit (DBU)** charge — the per-second fee Databricks itself charges for using the platform. It does **not** waive the underlying **Azure infrastructure** (the virtual machines, managed disks, and networking that a classic cluster actually runs on). Those are billed by Azure against your subscription/credits from the moment a cluster starts, trial or not. This is the single most common source of unexpected charges for beginners on this tier.

What this means in practice:

- Prefer **serverless** compute (notebooks, jobs, SQL warehouses) over a classic all-purpose cluster wherever this roadmap allows it — serverless bills only for actual execution time, with nothing left idling.
- If a task does need a classic cluster, set **Auto Termination to 10-15 minutes** every single time you create one — this is the #1 way beginners rack up silent charges.
- Use the smallest available node type and a single-node cluster for every task in this project — nothing here needs a multi-node cluster.
- The 14-day window is a DBU credit window, not a hard cutoff — nothing forces an automatic upgrade to a paid plan the moment it ends, but don't rely on that: check the workspace's "Manage trial" / pricing tier screen periodically so you know exactly where you stand.
- Check **Azure Cost Management + Billing** (Azure Portal → search "Cost Management") every couple of days while working through this project. Filter by resource group to isolate Databricks-related spend.

3. Business Scenario

E-commerce/Retail Data Platform. An online retail company sells products through its own storefront. It generates six core data streams: customers, products, orders, order_items, payments, and shipments. In a real company these would land from an operational database, a payment gateway, and a shipping API — for this project we generate small, realistic CSV files ourselves so nothing depends on external systems.

The Gold layer at the end of the pipeline answers concrete business questions: daily sales, best-selling products, customer lifetime spend, revenue by category, and orders by region — the tables a BI dashboard or analyst would query directly.

4. Target Architecture (built up gradually, not all at once)



We build this bottom-up: Task 1 starts with nothing more than `spark.read` on a single CSV. Storage, Auto Loader, orchestration, and triggers are each introduced only once the layer beneath them is solid.

5. Planned Catalog / Schema Structure (set up in Task 3)

Once Unity Catalog access is confirmed, the project will use one catalog with three schemas, matching the medallion layers:

```
ecommerce_dev (catalog)
|-- bronze (raw ingested tables: orders, customers, products, order_items, payments, shipments)
|-- silver (cleaned/standardized: orders_clean, customers_clean, ...)
|-- gold (business-ready: daily_sales, customer_summary, product_performance, revenue_by_category)
```

Exact catalog name and whether it's managed or external depends on what Task 0's remaining checks show — this will be finalized in Task 3.

6. Full Task Roadmap

We go through these one at a time in chat — building, running, and debugging each before moving to the next. This table is a reference map of where we're headed, not content to work through on your own.

Task	Focus	Difficulty
0	Inspect existing environment	Very Easy
1	Direct CSV load: spark.read to Delta table	Very Easy
2	Core transformations: select/filter/join/groupBy etc.	Easy
3	Unity Catalog: catalog / schema / table structure	Easy-Medium
4	Medallion architecture: Bronze / Silver / Gold	Medium
5	Azure Storage landing area (ADLS Gen2)	Medium
6	Ingestion through staging (multi-file)	Medium
7	Auto Loader: incremental ingestion	Medium-Hard
8	Incremental ETL: checkpoints, idempotency	Hard
9	Full pipeline: Storage -> Bronze -> Silver -> Gold	Hard
10	Lakeflow Jobs: orchestration, dependencies	Hard
11	Scheduling the workflow	Hard
12	Event-driven ETL: file-arrival triggers	Very Hard
13	Success/failure notifications	Very Hard
14	Data quality checks	Very Hard
15	Failure handling: intentional breakage + recovery	Very Hard
16	Final production-style project + deliverables	Very Hard

7. Cost-Control Checklist (applies to every task)

- Before creating any resource: confirm what it is, why it's needed, whether it can cost money, and how to delete it (this guide's per-task format always covers this).
- Auto-termination set to 10-15 min on any classic cluster.
- Prefer serverless compute wherever the task allows it.
- Stop/delete SQL warehouses and clusters at the end of every session, not just every task.
- Keep datasets small (a few hundred to a few thousand rows) — nothing in this roadmap needs large data to teach the concept.
- Check Azure Cost Management every few days, filtered to your resource group.
- Delete anything created purely for a mini-exercise once you've verified it worked.

8. How We Proceed From Here

This document is a reference, not a substitute for the hands-on flow you asked for. We still go task-by-task in chat: I give you one task's instructions, you build it in your workspace, send me your result or error, and only then do we

move on. Once the remaining Task 0 checks (Unity Catalog, storage, serverless availability) are confirmed, we start Task 1. I can regenerate this PDF later with your finished architecture diagram, final catalog structure, troubleshooting notes, and resume/interview material once the project is further along.